On the possibility of machinic comprehension

Agenda

1. **Introduction**
2. A problem I faced early in my NLP journey - how would you solve it?
3. Abstract language templates - why does this work?
4. Language as a natural encoding scheme
5. Cultural perspectives
6. The (im)possible task

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Context

On 12.10.2020 at about 11.45 A.M. the victim, since injured, [redacted] Mallick was travelling in a Motor cycle along the extreme left side (semi – metal portion of the road) of Amta Gagan more near Mahabir Nursing Home at a moderate speed . At the material time a Lorry bearing No. [redacted] was proceeding from " Uluberia Side " towards " Amta Side " at a terrific high speed, recklessly endangering human life and safety of others and knocked down from behind the victim alongwith his motor cycle with a great force and violating the Traffic/Road Rules. As a result, the victim sustained severe injury on his head, left leg with compound fracture and left hand , handicapped 60% which caused his permanent disability.

Rash and negligent driving on the part of the driver of the said vehicle bearing No. [redacted] (Lorry) was the sole cause of the pathetic accident which could easily be avoided if the reckless driver would take proper care and reasonable attention at the material time being devoid of ordinary sense of prudent driving. Police started a criminal case against the driver of the said vehicle bearing No. [redacted] U/S. 279/338 I. P. C.

The victim, since injured, was a smart , young and active man. He was the sole bread-winner of his family . He had all qualities to prosper his life in the near future. The accident made him permanent disablement. He sustained mental shock, pain and agony. The accident forces him to pass a life of miserable days.

I wish to claim compensation under section 166 of the Motor Vehicles Act. 1988 only.

OR

I/We wish to make a claim for compensation under the said Act and also in pursuance of the right on the principle of fault.

23. Any other information that
May be necessary Or helpful
In the disposal of the claim
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BERT + SpellCheck Approach (OCRfixr)

→ matching context-based suggestions from the BERT architecture with a spelling correction suggested on the basis of Levenshtein distance algorithms

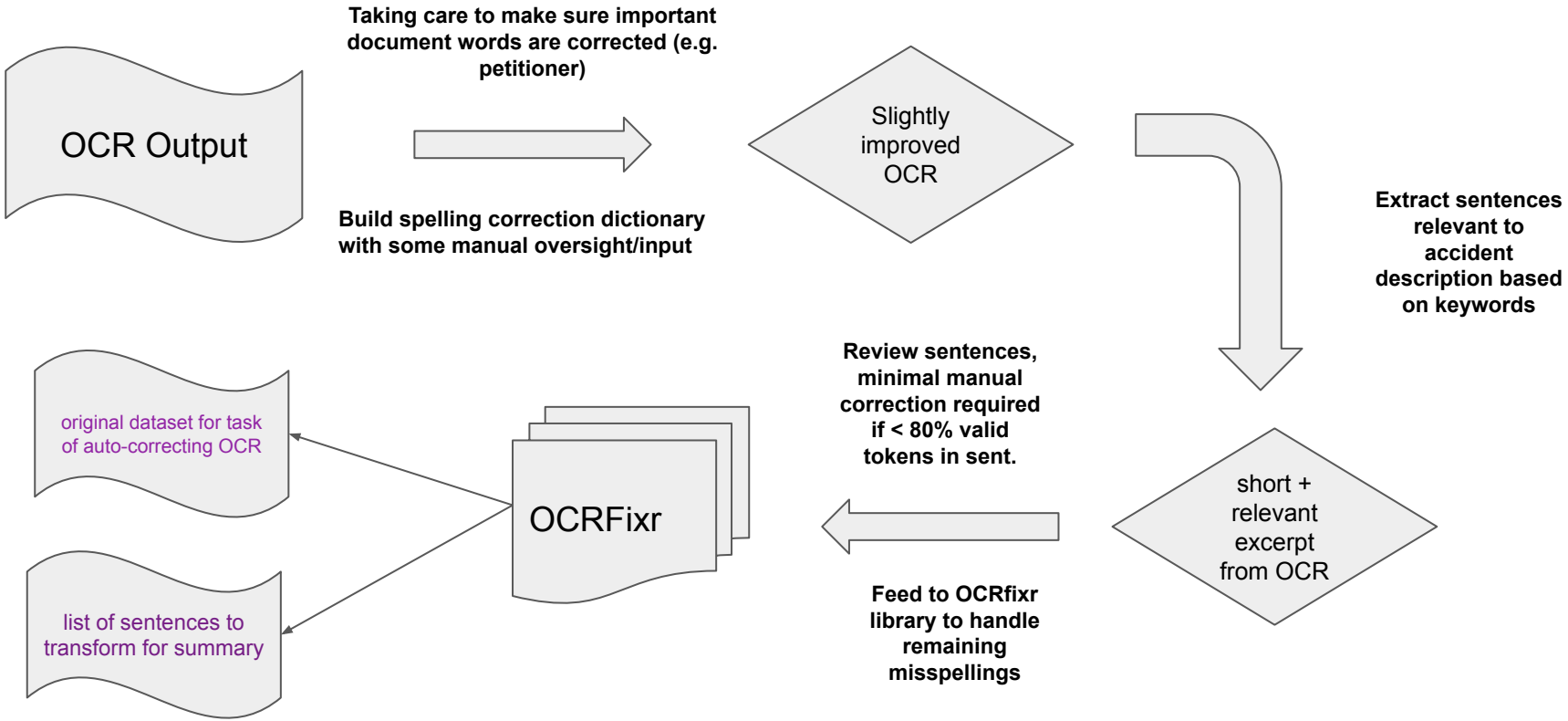
→ **Requirements:**

- meaningful context i.e. 80-90% of the words in sentences must be recognisable to BERT at minimum, this will require manual labour

→ **Pros:**

+ approach is already implemented, with adjustable features in OCRfixr library
+ manual labour required will help build a dataset to train a seq2seq model that can entirely automate this approach in the future for new data

BERT + SpellCheck Roadmap



BOW + Generative Approach (word2vec + k2t/CGMH/sG)

→ foregoing most manual labour (with the exception of constructing common misspellings dict w/ SpellCheck), ignoring invalid tokens in a sentence and treating it as a bag-of-(key)words to create vector embeddings

→ the subsequent task is forming a cohesive sentence from these valid words

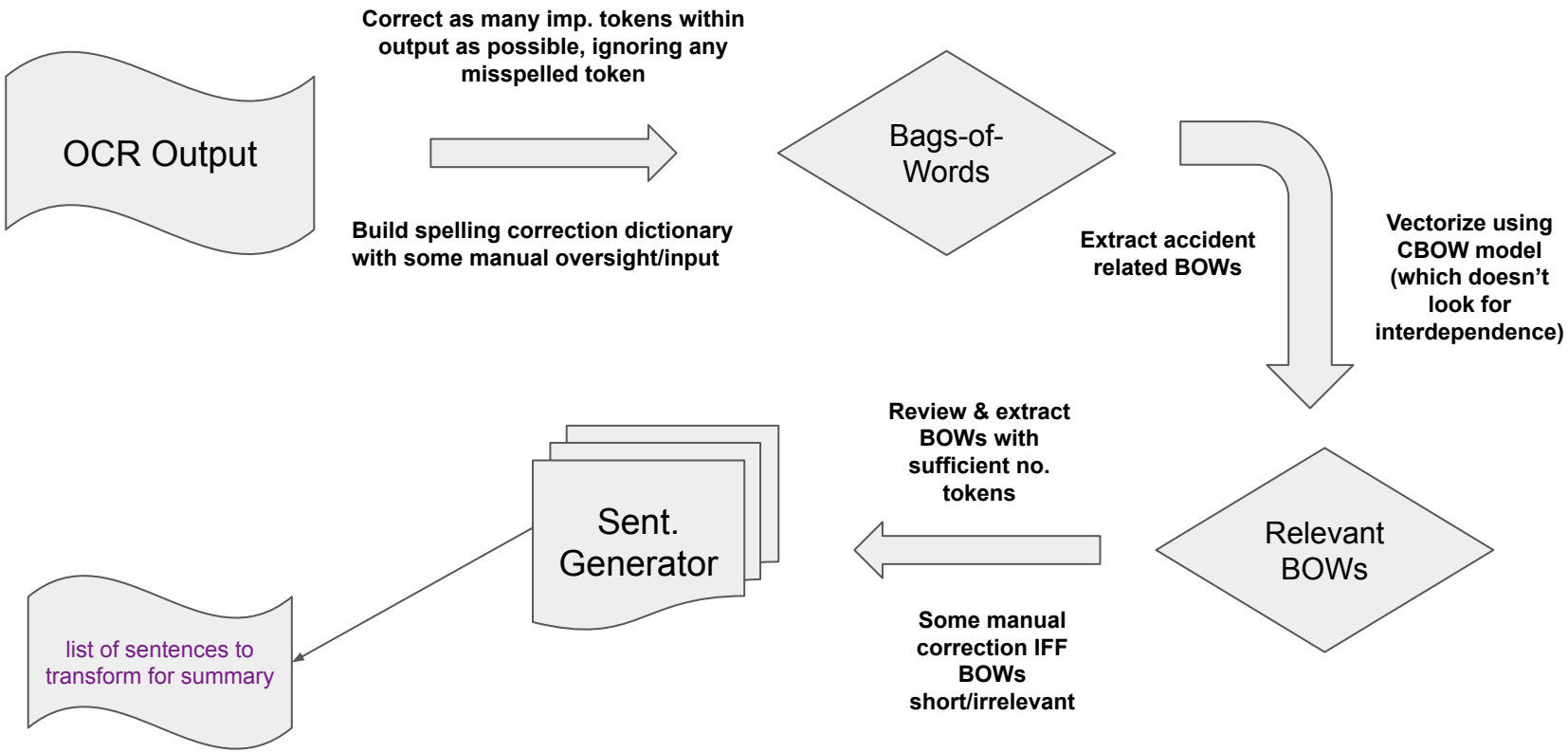
→ **Requirements:**

- bags of words need to be somewhat meaningful, converting <4 tokens into a logical sentence will be a difficult task and not perform well in praxis

→ **Pros:**

+ if, after applying corrections from constructed SpellCheck dict, the sentences are bags of words with short length, this issue can be mitigated by additional manual correction of tokens
+ sentence generators guarantee syntactically valid sentences as output, which will be easy to summarize

BOW + Generative Roadmap



Results

What would you do?

What did I do?

Analytical Template + Extraction Approach

→ manual analysis of documents to determine information about accidents that are (for the most part) consistent between all OCR outputs

→ with knowledge of this, it is possible to create a template for summarisation which would require information extraction to fill in the “blanks”

- e.g. The accident took place in <proper noun>. The petitioner was <verb> when the offender <predicate>. This resulted in <predicate>.

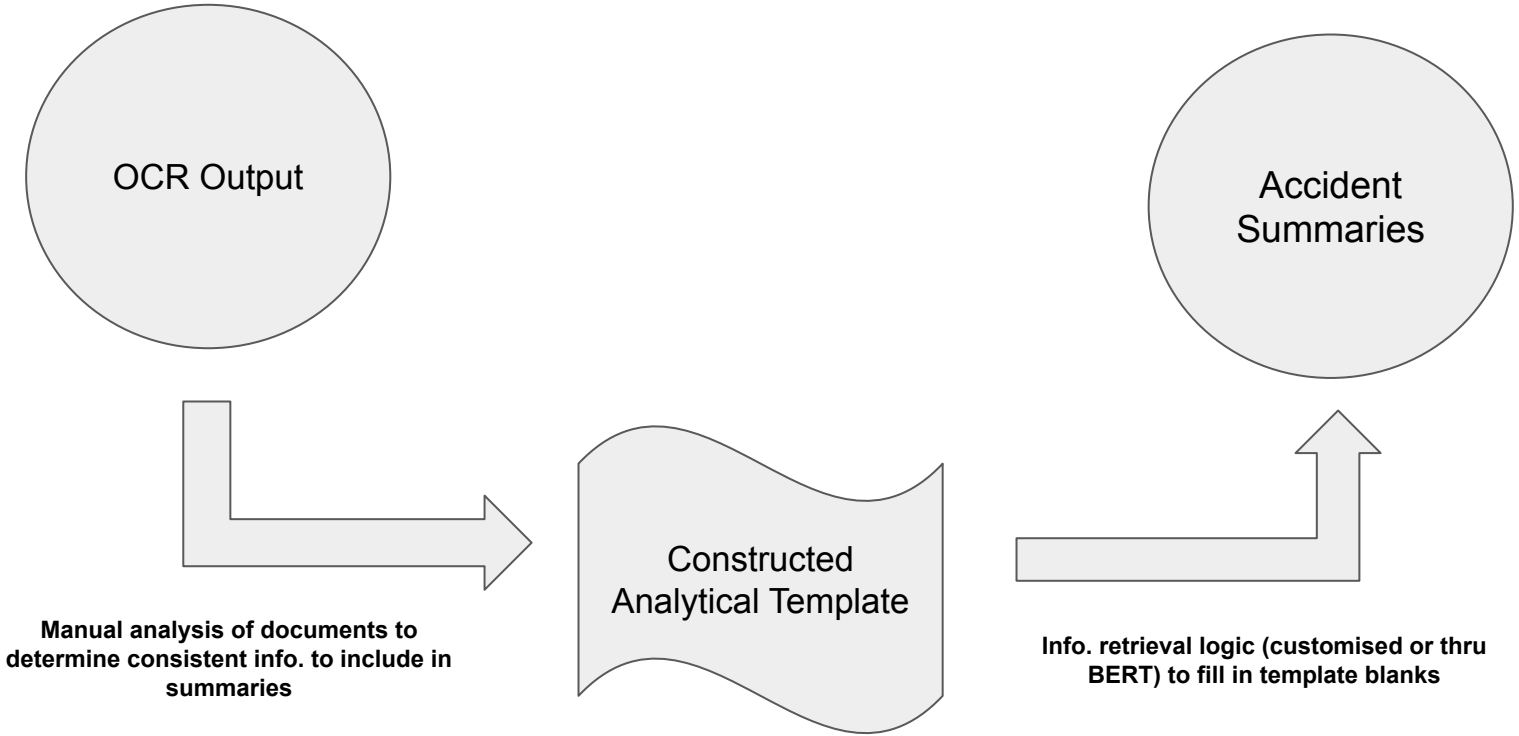
→ **Requirements:**

- manual analysis of all documents to find a suitable template for summarisation
- robust logic to handle extraction of relevant clauses to fill in blanks

→ **Pros:**

+ eliminates the need for tedious correction of OCR output beyond initial common corrections
+ relatively fast hit-or-miss approach i.e. we will know if this works well immediately b/c there are less variables

Analytical Template + Extractive Roadmap



Results

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A mass of templates in natural language

plastic knife

paper cup

cheese cake

silver spoon

wooden stick

bronze statue

silicon chip

glass table

sand castle

brick house

Attribute x Artifact

We don't "understand" things bottom-up; by recognising patterns after seeing them n times

Instead, "understanding" is a combination of pattern recognition and verification of intuition

Understanding v.s. Recognition

- NLP has achieved the state-of-the-art (or come close to its limit) in tasks of recognition
- “Understanding” of an arbitrary clause requires the knowledge of a universe/model where said arbitrary clause can be verified in truth
  - **On the basis of necessary and sufficient conditions**

$$\text{“Rudolph is a reindeer”} \rightarrow \text{REINDEER(“Rudolph”)}$$

$$\text{“Arabella is an activist”} \rightarrow ?$$

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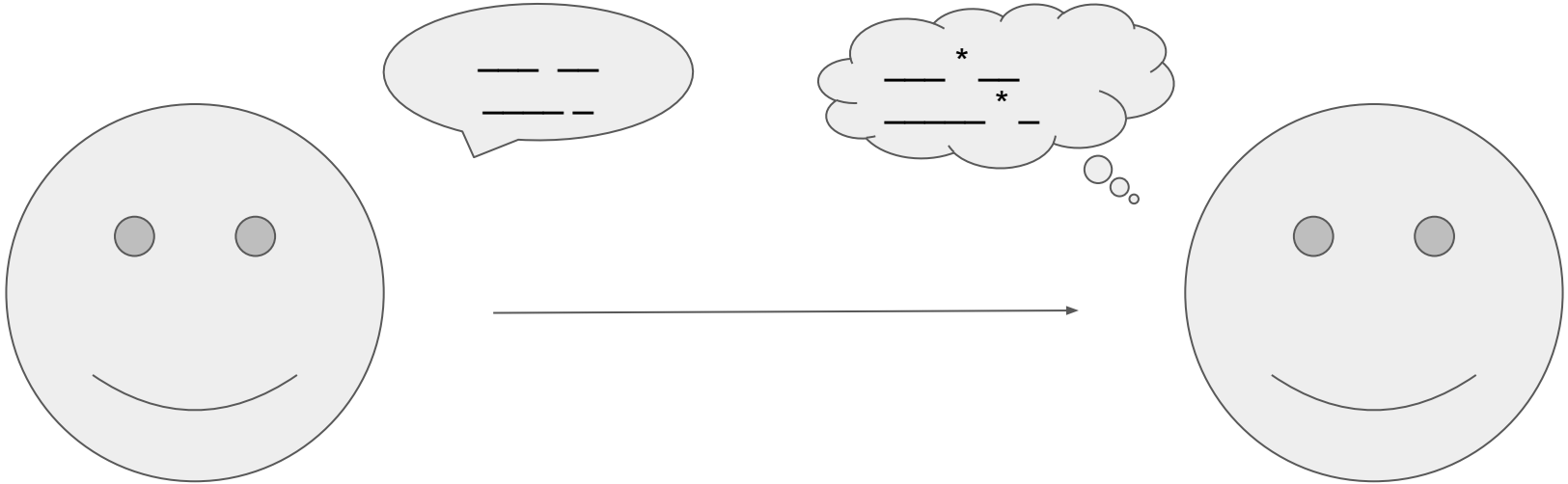
Ambiguity as an Encoding Scheme



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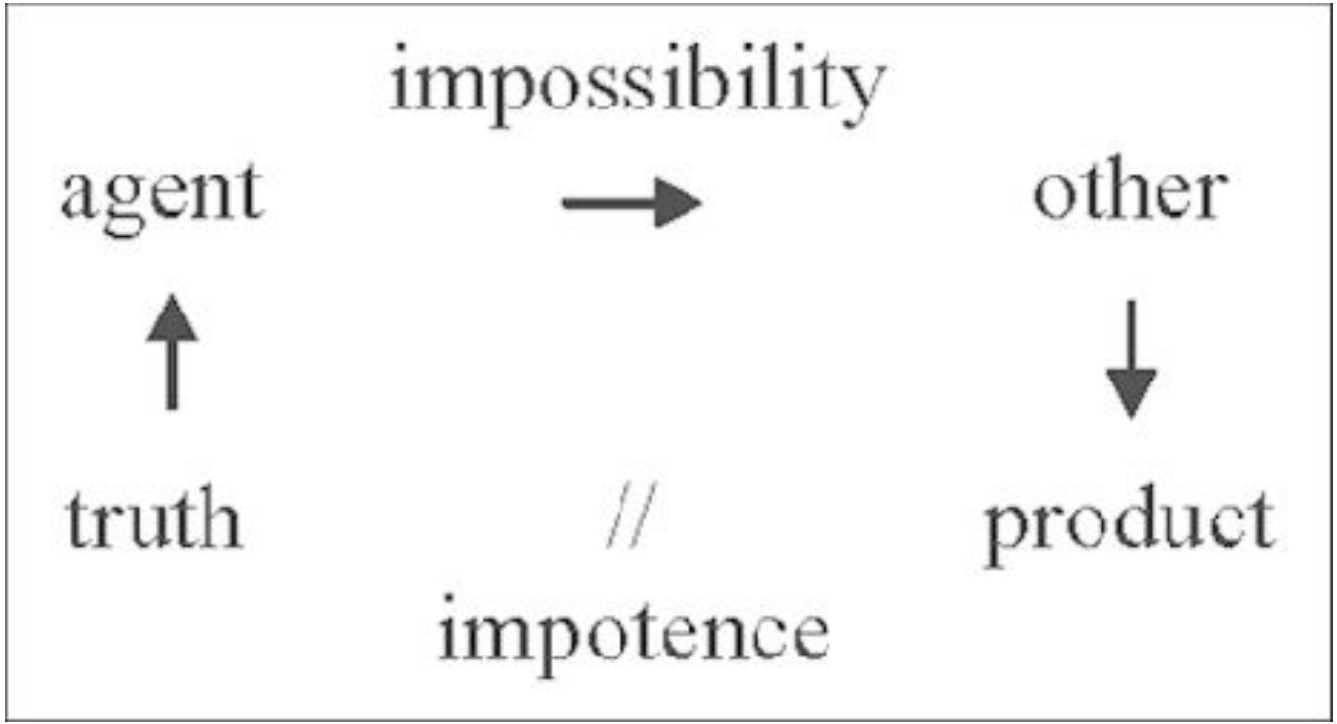
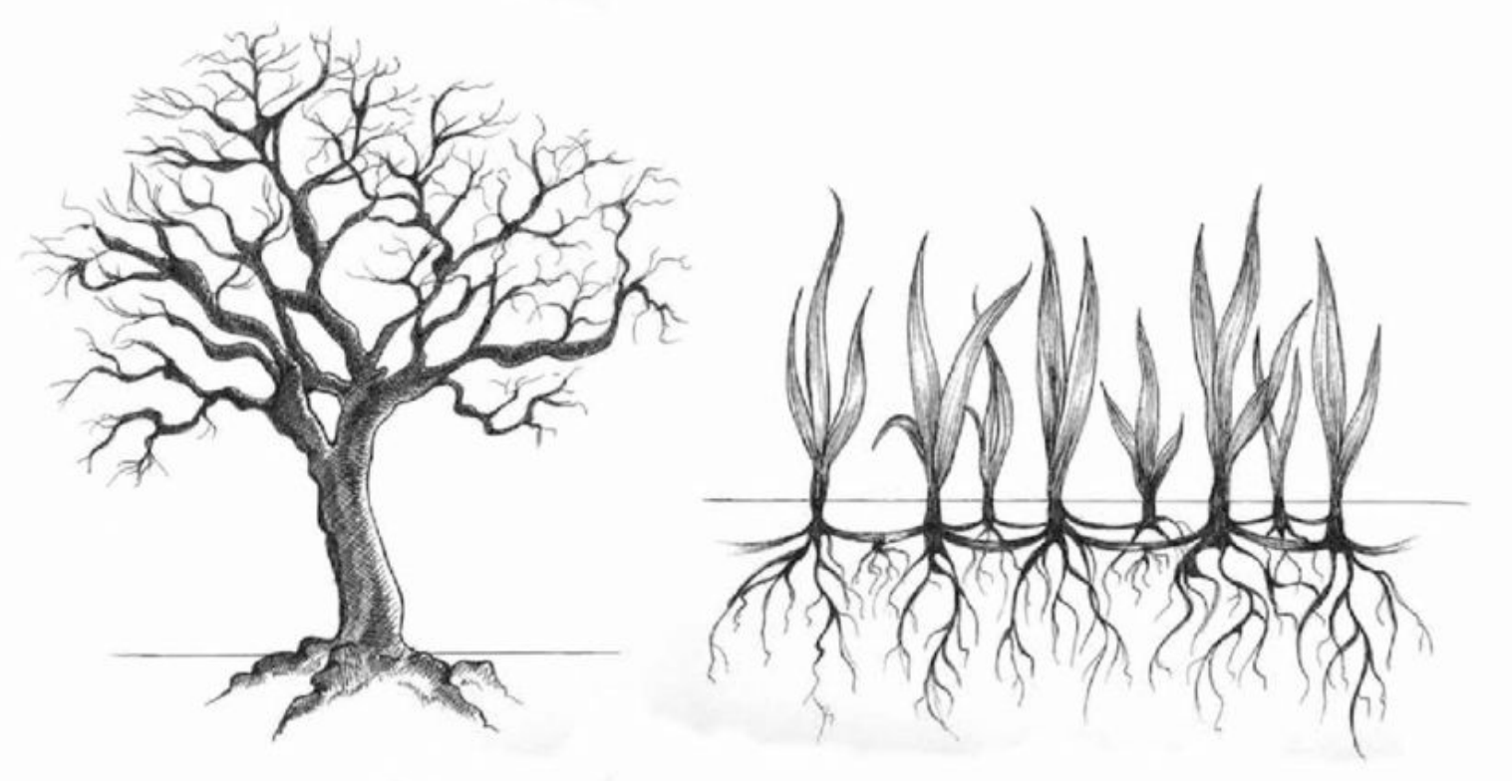
Lacan on Language



Expanding on Lacan’s Truth - Deleuze on Territorialization



So why does this matter?

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(Im)possibility of Machine Comprehension

- **Natural language is already “encoded” - it can’t be analysed literally**
- **Machines lack a world model whereby knowledge can be grasped, verified**
- **“Filling in the gaps” is a complex phenomenon innate to communication**
- **Natural language ≠ formal defined language; there is no type hierarchy that encompasses a universal set of cognitive primitives**
  - **Existence of a universal grammar has not been sufficiently proved**
  - **Language is constantly changing with culture; how it is used and understood**

So what can we do?

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